

Roll No.

TMS-301

**B. TECH. (R & A/AME) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

PRODUCT DESIGN AND DEVELOPMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the importance of product specification with respect to product success using suitable example. (CO2, CO1)
- (b) What do you understand by multi-functional product ? (CO2, CO1)
- (c) Describe the importance of management in industrial design processes. (CO2, CO1)
2. (a) What are the trends of product decision in 'Product Design and Development' ? (CO2)

P. T. O.

(2)

- (b) Define Stage Gate process of product development with suitable flow diagram. (CO2)
- (c) Discuss the concept of design for safety and environmental considerations in product design. (CO3)
3. (a) Explain the role of Inventions and Innovation in product development. (CO1, CO2)
- (b) Explain "Creativity" with respect to new product development. (CO1, CO2)
- (c) What is 'Ergonomics'? Explain the importance of ergonomics in Automobile. (CO2, CO3)
4. (a) What do you understand by "Refining the Product Specification"? Explain with examples. (CO3)
- (b) What is perceived quality? Also make flow chart to explain the perceived quality. (CO3)
- (c) Discuss the role of value analysis in product design. (CO4)
5. (a) What are the different success criteria of product development? Explain. (CO4, CO5)
- (b) What are the factors for material selection? Describe the BOM importance in production. (CO4, CO5)
- (c) What is Product Life Cycle Management (PLM)? (CO6)

Roll No.

TRA-301

B. TECH. (ME) ROBOTICS (THIRD SEMESTER)

END SEMESTER EXAMINATION, 2022

MICROPROCESSOR AND MICROCONTROLLER

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Represent decimal values (-6) , (-24) , (-19) , (-42) , and (-31) , as signed, 7-bit numbers in the following binary formats : (CO1)

(i) Signed Magnitude

(ii) 1's-complement

(iii) 2's-complement

(b) Represent each of the following decimal numbers in Binary, Octal, and Hexadecimal number systems : (CO1)

(i) 128

P. T. O.

(ii) 365.324

(iii) 134

(iv) 210.136

(v) 456

(c) (i) $\text{ADD } (6B7D)_{16} + (F33B)_{16}$

(CO1)

(ii) Subtract $(10001001)_2 - (01011100)_2$

(iii) Multiply $(4A6C)_{16} \times (D0)_{10}$

(iv) Multiply $(7214)_8 \times (05)_8$

2. (a) Explain the detail architecture of 8085 microprocessor.

(CO2)

(b) Explain the following pins :

(CO2)

(i) ALE

(ii) TRAP

(iii) RESET

(iv) A_8 to A_{16}

(v) S_1 and S_0

(c) Explain the various registers used in 8085 microprocessor.

(CO2)

3. (a) Write an assembly language program to add two 16 bit numbers with and without carry.

(CO2, CO3)

(b) Interface a 8085 microprocessor based system with 4kB RAM whose starting address is (0000) H.

(CO2, CO3)

- (c) Differentiate between peripheral mapped I/O and memory mapped I/O interfacing. (CO2, CO3)

4. (a) Explain the various methods to specify the operands and explain with examples for each. Mention the method applied for instruction : (CO4)

(i) SHLD 2500H

(ii) MOV A, M

(iii) MVI A, 67H

- (b) Explain the PUSH and POP Instructions. Explain the following codes and the content of stack pointer after the execution : (CO4)

2000 LXI SP, 3099H

2003 LXI H, FFFF H

2006 PUSH H

2007 MOV A, B

2008 MOV C, D

2009 POP H

200A HLT

- (c) Write an assembly language program to find largest number in an array of numbers. (CO4)

5. (a) Differentiate between Microprocessor and Microcontroller.

(CO5 & CO6)

- (B) Explain the Block diagram of microcontroller 8051 with the registers used. (CO5 & CO6)

P. T. O.

- (c) Draw and explain the PSW of 8051 microcontroller. Analyze the code and explain each line of the code given below : (CO5 & CO6)

MOV A, #-128

MOV R1, #-4

ADD A, R1

RET

Roll No.

TAE-301

**B. TECH. (AME) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

AUTOMOTIVE SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What exactly do you mean by an automobile ? How do they contribute to economic growth ? (CO1, CO3, CO4)
- (b) Name the top *three* German automakers and list their automobile variants. (CO1, CO3, CO4)
- (c) What do you understand by "special purpose vehicle" ? List various special purpose vehicles and their intended uses. (CO1, CO3, CO4)
2. (a) Prepare a table of the electrical and electronic installation of an automobile, mentioning its purpose of use. (CO2, CO4, CO5)

P. T. O.

(2)

- (b) Draw a constant mesh gearbox and explain how it works with gear shifting. (CO2, CO4, CO5)
- (c) Draw a typical hydraulic braking system and explain how it works. (CO2, CO4, CO5)
3. (a) What are the various aerodynamic forces that act on an automobile at high vehicle speeds ? Explain any *one* in detail. (CO3, CO5, CO6)
- (b) Share your thoughts on electric vehicles. Are they the future of modern automobiles ? (CO3, CO5, CO6)
- (c) What are the various factors that affect the architecture of an automobile ? Explain. (CO3, CO5, CO6)
4. (a) Categorise the systems and subsystems of an automobile. Give an overview of the major systems in an automobile. (CO1, CO2, CO5)
- (b) Write short notes on plug-in hybrids and hydrogen fuel cells. (CO1, CO2, CO5)
- (c) How do electric vehicle transmissions differ from those of conventional automobiles ? Explain. (CO1, CO2, CO5)
5. (a) How do teardown and benchmarking aid in the rapid release of automobiles ? Explain. (CO1, CO2, CO5, CO6)
- (b) Name the various teardown and benchmarking processes. Explain any *one* in detail. (CO1, CO2, CO5, CO6)
- (c) What are the various roles and responsibilities of a teardown and benchmarking engineer ? (CO1, CO2, CO5, CO6)

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TME-302

**B. TECH. (ME) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

MATERIAL SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Derive an expression for the A.P.F. in hcp structure. (CO1)
(b) Draw the sketches for different Bravais lattices indicating their angular and linear dimensions. (CO1)
(c) What is a grain boundary ? Differentiate between line defects and point defects. (CO1)
2. (a) Explain stress strain curve of a mild steel specimen with a neat and labelled sketch. (CO2)
(b) Define the term fatigue. What are the methods used for improving the fatigue life of the component ? (CO2)
(c) Explain the mechanism of failure in ductile materials. (CO2)

P. T. O.

(2)

3. (a) Draw a neat and labelled diagram of T.T.T. curve and explain the importance of various stages involved in it. (CO3)
(b) Explain the iron carbon equilibrium diagram clearly mentioning all the phases and the reactions involved in it. (CO3)
(c) Explain a binary phase equilibrium diagram having complete liquid solubility and incomplete solid solubility with the help of a neat and labelled diagram. (CO3)
4. (a) What is the purpose of heat treatment ? Draw the diagram for mentioning the heat treatment processes and their ranges with respect to the temperature and carbon percentage. (CO4)
(b) Explain the Jominy End Quench test. (CO4)
(c) Differentiate between the nitriding and cyaniding heat treatment process. (CO4)
5. (a) Classify the different types of steels on the basis of their carbon percentage, properties and applications. (CO5)
(b) Define corrosion and the methods to prevent corrosion. (CO5)
(c) Mention the applications of ceramics and polymers. Write a short note on composites. (CO5)

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TMA-303

B. TECH. (ME) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023

ENGINEERING MATHEMATICS—III

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Show that the function $f(z) = u + iv$, where

$$f(z) = \begin{cases} \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases} \text{ is not analytic at } z = 0, \text{ although}$$

Cauchy-Riemann equations are satisfied at $z = 0$. (CO1)

P. T. O.

- (b) If $f(z)$ is a harmonic function of z , then show that : (CO1)

$$\left\{ \frac{\partial}{\partial x} |f(z)| \right\}^2 + \left\{ \frac{\partial}{\partial y} |f(z)| \right\}^2 = |f'(z)|^2$$

- (c) Find the values of α and β such that the function : (CO1)

$$f(z) = (x^2 + \alpha y^2 - 2xy) + i(\beta x^2 - y^2 + 2xy)$$

is analytic. Also find $f'(z)$ in terms of x and y .

2. (a) Use Cauchy's integral formula to evaluate $\int_C \frac{z^2 + 23}{(z-2)(z-3)} dz$, where

C is the circle $|z - 4i| = 3$. (CO2/CO3)

- (b) Evaluate the complex integral $\int_{1+i}^{2+4i} (x^2 + ixy) dz$ along the curve

$x = t$ and $y = t^2$. (CO2/CO3)

- (c) Find the bilinear transformation which maps the upper half of the z -plane into the unit circle in the w -plane in such a way that $z = i$ is mapped into $w = 0$ while the point at infinity is mapped into $w = -1$.

(CO2/CO3)

3. (a) Using Newton's method, find a real root of equation $3x - 1 = \cos x$ correct to three decimal places.

(CO4)

- (b) Using Regula-Falsi method to compute a real root of the equation $x = e^{-x}$ correct to three decimal places.

(CO4)

- (c) A car is moving at the speed of 30 m/sec. Suddenly brakes are applied. The speed of the car per second after 't' seconds is given below : (CO4)

Time (t)	Speed (v)
0	30
5	24
10	19
15	16
20	13
25	11
30	10
35	8
40	7
45	5

Using Simpson's $\frac{3}{8}$ rule to compute the distance moved by the car in 45 seconds.

4. (a) Solve $\frac{dy}{dx} = -xy$ numerically from $X = 0$ to $x = 1$ using explicit forward first order Euler method with initial condition of $y(0) = 1$ and step size (h) of 0.2. Find the absolute value of error in $y(1)$ calculated using analytical and numerical solution in percentage (%) (calculate the error using analytical solution as the basis and use three decimal places).

(CO5)

- (b) Using Runge-Kutta method to obtain y for $x = 0.2$ correct to three decimal places. Given that the initial value problem (IVP) $\frac{dy}{dx} = x + y$ with initial condition $y(0) = 1$.

(CO5)

P. T. O.

- (c) Consider an initial value problem (IVP) : $\frac{du}{dt} = t + u + tu$ with $u(0) = 1$. Given IVP solved by numerically using Taylor's method, calculate $u(0.1)$ taking step-size ($h = 0.5$). (CO5)

5. (a) The number of airplane landings at an airport in a minute (X) and their probabilities are given in the following table : (CO6)

X_i	$P(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Find the mean and standard deviation of X.

- (b) If the probability density function of a continuous random variable X is given by :

$$f(x) = \begin{cases} kx^2; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the value of k and hence compute its mean, variance and standard deviation. (CO6)

- (c) The distribution of the number of road accidents per day in a city is Poisson with mean 4. Find the number of days out of 100 days when there will be (i) no accident, (ii) at least 2 accidents. (CO6)

Roll No.

TME-304

B. TECH. (ME) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023

BASIC THERMODYNAMICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

(v) Make suitable assumptions for any missing data.

1. (a) Explain the following with suitable examples of each : (CO1)

(i) Processes and cycles

(ii) Path and point functions

(iii) Thermodynamic equilibrium

P. T. O.

(b) Complete the following table based on the first law of thermodynamics :

(CO3)

	Q_{in}, kJ	W_{out}, kJ	U_1, kJ	U_2, kJ	m, kg	$u_2 - u_1 \text{ kJ/kg}$
(i)	600		850	600	2	
(ii)	-250	-200	750		5	
(iii)		100		50	4	-150
(iv)	-400		30	120	4	
(v)		500	200	450		100

Note : These are individual processes and do not constitute a cycle.

(c) Sketch the P-v, T-s and P-T diagram of water and explain the salient points. (CO1)

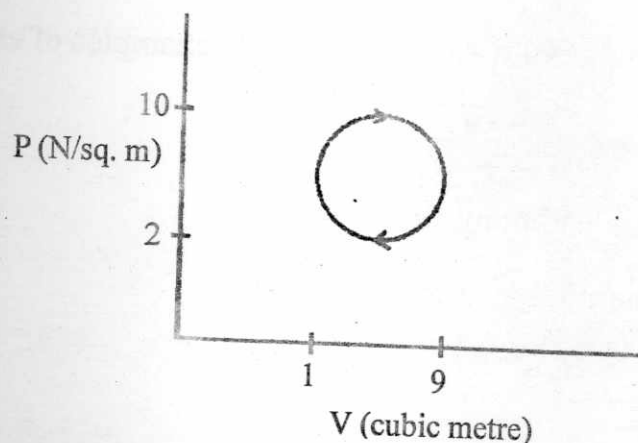
2. (a) Derive an expression for the increase in temperature inside an empty gas cylinder as it is filled from the pipeline. (CO4)

(b) For the process shown below, calculate : (CO2)

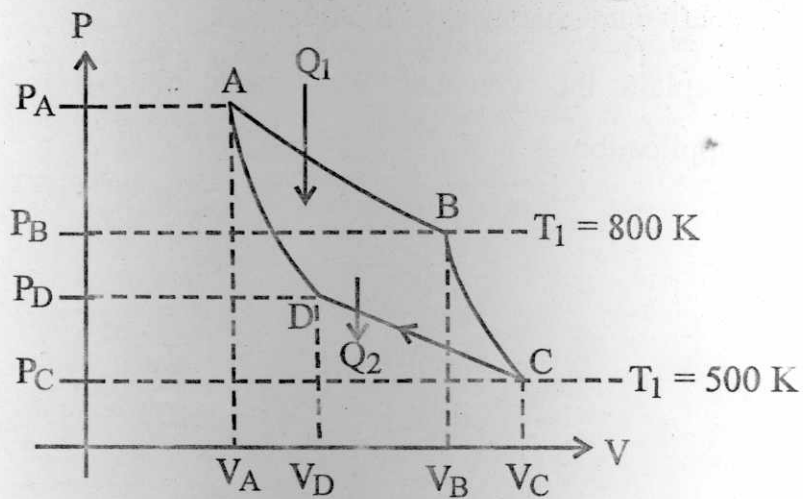
(i) W

(ii) ΔU

(iii) Q



- (c) Explain the concept of thermodynamic temperature scale. (CO2)
3. (a) Describe the Carnot cycle and derive an expression for its efficiency. (CO4)
- (b) The PV diagram of a heat engine is shown below. The heat input is 7000 J. (CO4)



P-V Diagram of Carnot Cycle

Find the efficiency, work output, heat rejected and draw the Ts diagram. Also explain all the processes.

- (c) Derive expressions for the efficiency of heat engines and COP of heat pumps. (CO4)
4. (a) Explain the Clausius inequality and prove the same. (CO5)
- (b) Air enters an adiabatic compressor at 300 K, 1 atm pressure. The work input to the compressor is 350 kJ per kg of air flowing through the compressor. Given that the compressor has an isentropic efficiency of

0.85, calculate the actual exit temperature and pressure of the compressor. Use the constant specific heat relations for this problem.

(CO5)

(c) Derive *two* Tds relations.

(CO5)

5. (a) Compare Dalton's and Amagat's Laws.

(CO6)

(b) Derive any *two* Maxwell's relations.

(CO6)

(c) Explain the Van der Waal's equation of real gas and also give applications.

(CO6)

Roll No.

TME-305

**B. TECH. (ME) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

MANUFACTURING PROCESS—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is pressure die casting ? Discuss with neat sketch. (CO1, CO2)
(b) What is the probable reason of defects in casting ? How can it be avoided ?
(CO1, CO2)
(c) What is the role of temperature in and manufacturing process ? How does it affect the external force and deformation of work material ?
(CO1, CO2)
2. (a) Write a short note on flow stress with suitable equation. (CO1, 2)

P. T. O.

(2)

- (b) What is the difference between true stress and engineering stress ?
Explain with the help of stress-strain curve. (CO1, CO2)
- (c) What is phenomenon of forging ? What is the mechanism of strain/work hardening in forging ? (CO1, CO2)
3. (a) Explain the following terminology used in rolling : (i) Length of contact, (ii) Roll pressure, (iii) Work done in rolling a strip with necessary equation. (CO3, CO4)
- (b) Differentiate between any *five* rolling and extrusion. . (CO3, CO4)
- (c) How is the blanking process different from piercing ? Elaborate. (CO3, CO4)
4. (a) Write any *four* differences between drawing and deep drawing. Give any *two* examples also. (CO3, CO 4)
- (b) Briefly explain the process of injection moulding with neat diagram. (CO3, CO4)
- (c) What is the importance of friction in the process of extrusion ? How is the heat generated in the process controlled ? Elaborate. (CO3, CO4)
5. (a) Explain in detail the process of powder metallurgy. (CO4, CO5)
- (b) What is rapid sintering ? What is its importance in powder metallurgy ? (CO4, CO5)
- (c) What is the role of binders used in powder metallurgy ? Name any *three* types of binders. (CO4, CO5)

TME-306

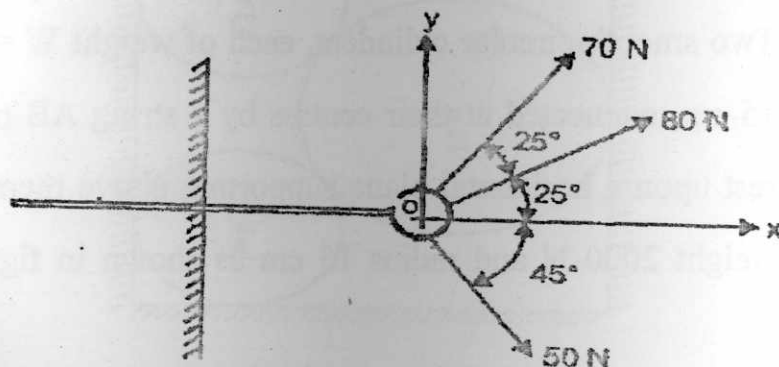
**B. TECH. (MECHANICAL ENGINEERING)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
ENGINEERING MECHANICS**

Time : Three Hours

Maximum Marks : 100

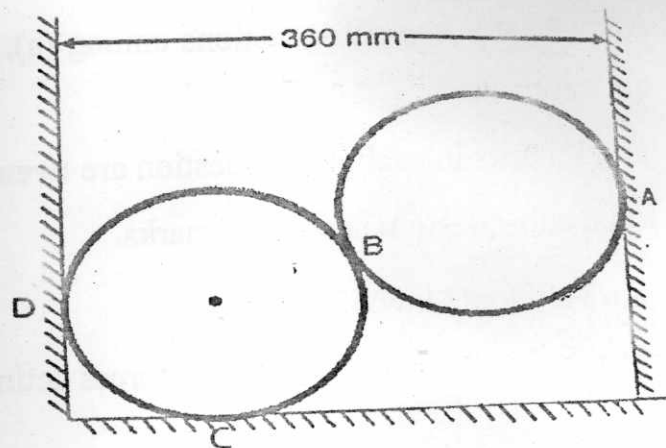
- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each sub-question carries 10 marks.

1. (a) Explain Newton's law of motions. (CO1, CO2)
(b) Determine the resultant of the three forces acting on a hook as shown in Fig. (CO1, CO2)



P. T. O.

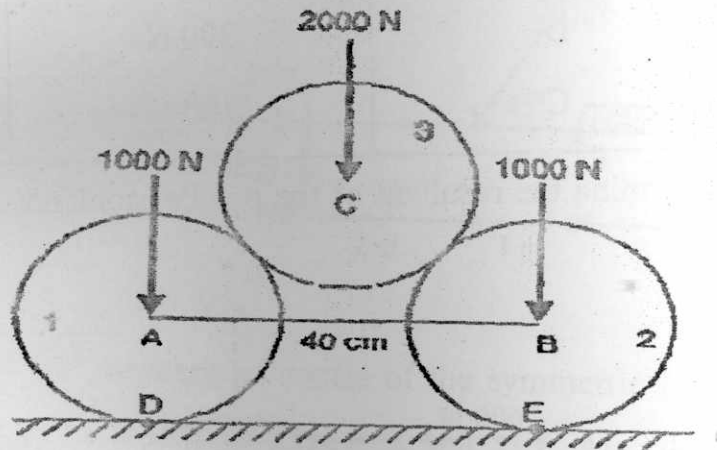
- (c) The resultant of two forces one of which is 3 times the other is 300 N. When the direction of smaller force is reversed, the resultant is 200 N. Determine the two forces and the angle between them. (CO1, CO2)
2. (a) Define a Free Body Diagram. Give two examples. State the conditions for equilibrium of a rigid body in three dimensions. (CO2)
- (b) Two smooth spheres each of radius 100 mm and weighing 100 N, rest in a horizontal channel having vertical walls, the distance between which is 360 mm. Find the reactions at the point of contact A, B, C and D as shown in Fig. (CO2)



- (c) Two smooth circular cylinders, each of weight $W = 1000$ N and radius 15 cm, connected at their centres by a string AB of length 40 cm and rest upon a horizontal plane supporting above them a third cylinder of weight 2000 N and radius 15 cm as shown in figure ahead. Find the

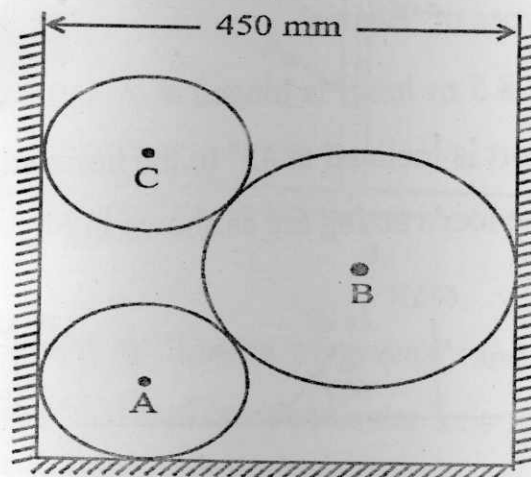
force in string AB and the pressure produced on the floor at the points of contact D and E.

(c) The resultant of two forces one of which is 3 times the other. (CO2)



3. (a) What do you understand by the term 'Couple' ? Discuss the characteristics of a couple. (CO2, CO3)
- (b) Three cylinders are placed in a rectangular ditch as shown in Fig. Neglecting friction, determine the reaction between cylinder A and vertical wall. Weights and radii of the cylinder are as given below :

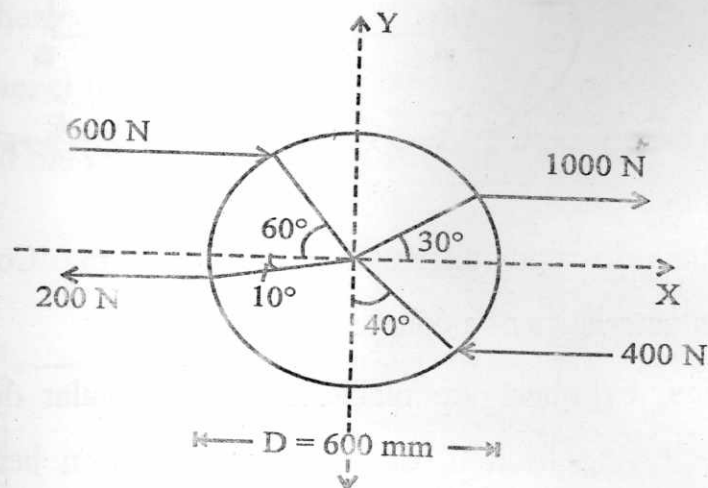
(CO2, CO3)



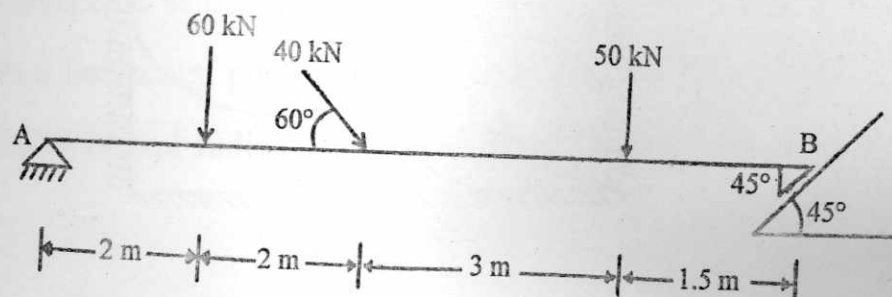
P. T. O.

Cylinder	Weight	Radius
A	75 N	100 mm
B	200 N	150 mm
C	100 N	125 mm

- (c) Determine the resultant of the parallel coplanar force system shown in Fig. (CO2, CO3)

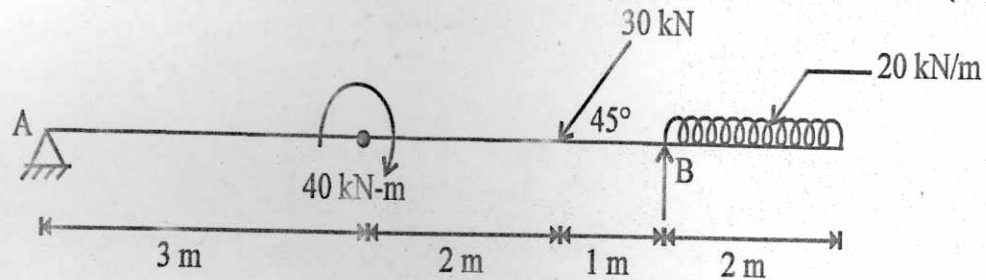


4. (a) What do you mean by 'Beam'? Explain different types of 'Supports' and the reactions that can be developed by the support. Also explain different types of 'Beams'. (CO4)
- (b) Beam AB, 8.5 m long, is hinged at A and has roller support at B. The roller support is inclined at 45° to the horizontal. Find the reactions at A and B, if the loads acting are as shown in Fig. (CO4)

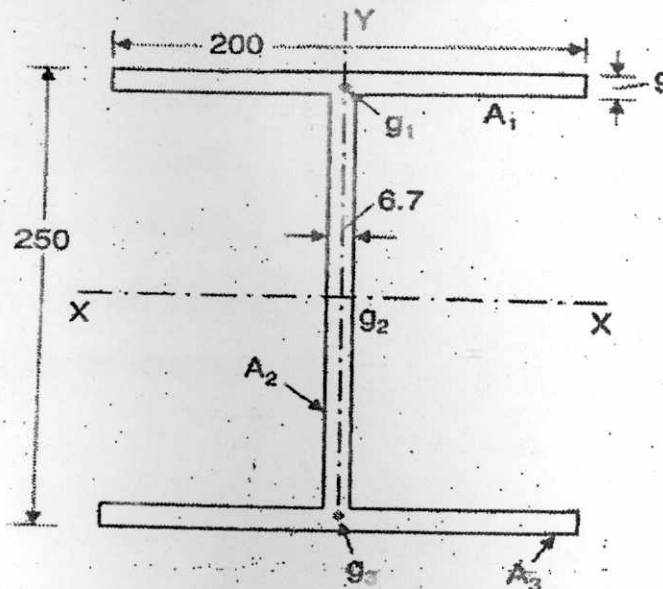


- (c) Determine reactions developed at supports in the beam as shown in Fig.

or circular hole cut in a thin plate so that this point will (CO4)



5. (a) Determine the moment of inertia of the symmetric I-section shown in figure about its centroidal axes X-X and Y-Y. Also determine moment of inertia of the sectional about a centroid axis perpendicular to X-X axis and Y-Y axis. (All dimensions are in mm.) (CO5)

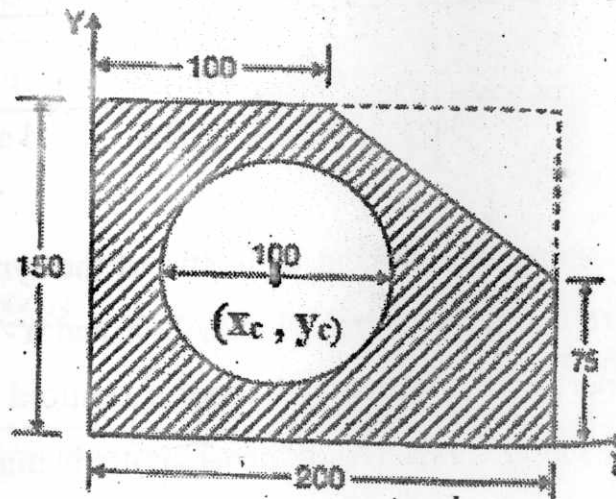


- (b) What do you mean by 'Centre of gravity' and 'Centroid'? Explain 'parallel axis theorem' and 'perpendicular axis theorem' of moment of inertia. (CO5)

P. T. O.

- (c) Determine the coordinates X_c and Y_c of the centre of 100 mm diameter of circular hole cut in a thin plate so that this point will be the centroid of the remaining shaded area shown in figure. (All dimensions are in mm.)

(CO5)



Roll No.

TME-307

B. TECH. (ME) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023

RENEWABLE ENERGY SOURCES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are the various sources of "Renewable Energy" ? Explain in brief.
(CO1/CO2)
- (b) Enumerate the criteria based on which energy can be classified. Give their availability, relative merits, demerits and classification.
(CO1/CO2)
- (c) Describe the various non-conventional sources relevant to India with focus on latest trends and global energy scenario. (CO1/CO2)
2. (a) Enumerate wind recording methods accepted by World Meteorological Organization. (CO3)
- (b) What are the basic components of wind energy conversion systems (WECS) ? Explain them with a neat block diagram. (CO3)

P. T. O.

(2)

- (c) Explain major factors which govern the selection of site for a proposed wind turbine generating system. (CO3)
3. (a) What are the fuel properties of biogas ? Explain briefly the methods used to obtain energy from biomass. (CO4)
- (b) How are Biogas plants classified ? Explain. (CO4)
- (c) Explain with a schematic diagram the "Anaerobic digestion system". (CO4)
4. (a) Classify tidal power plants. Explain briefly various components of a tidal power plant. (CO5)
- (b) Classify geothermal resources. Explain geothermal system with a neat sketch. (CO5)
- (c) Enumerate applications of geothermal energy. List down advantages and disadvantages of geothermal energy over other energy forms. (CO5)
5. (a) Analyze the working of thermoelectric generators and MHD power generation. (CO6)
- (b) What is "Direct energy conversion device ? Compare thermoelectric generator and thermionic converter. (CO6)
- (c) Discuss and explain the principles of energy conversion. (CO6)